The lesion is dosage insufficiency, not a druggable target

Compound heterozygous BUB1B

p.Leu737Ter — deletes the pseudokinase domain

p.Asn1002Lys — buried inside it, 20% accessible

You cannot inhibit your way out of an insufficiency.

No target on BubR1 — and one point we decline to claim

Vertebrate BubR1 is a pseudokinase — dead, but essential scaffold

Missense sits 34.7 Å from the nucleotide site

`Open Targets: 'no druggable pocket'  —  WE DO NOT USE THIS.`

That annotation needs an experimental structure.

None covers this domain. Absence of evaluation, not a negative.

What we would do first is not a drug

MVA predisposes to Wilms tumour, including bilateral

This child is already under renal follow-up (nephrocalcinosis)

→ Serial renal ultrasound.

```
Near-zero incremental cost. Zero risk.

Better benefit-to-risk than any drug in this report.
```

Lead drug proposal: mTOR inhibition — two chains converge

```
Oncological:   temsirolimus, randomised trials in rhabdomyosarcoma
               his histology, his age, added to chemotherapy
Mechanistic:   mTORC1 hyperactive in BubR1-mutant mice — his genotype
```

Our own objection: he has muscle atrophy, short stature, failure to thrive. mTORC1 drives growth. The mouse rationale and the clinical reality oppose each other on the same tissue.

Readthrough: right codon, wrong product

Stop created = TGA-A — the most permissive class

UGA readthrough installs Trp, Cys or Arg — not Leu

Position 737 across 247 vertebrate orthologs:

```
Leu 236      Trp 0     Cys 0     Arg 0
```

But Ser (-77.7 Å³, 5 spp) is a BIGGER cavity than Cys (-58.2).

Only Trp exceeds the tolerated range. We demoted it to third.

What we argue against — and what this is

Aneuploid tumours depend on residual checkpoint, so checkpoint inhibitors are the standard play. NOT HERE: his normal cells are checkpoint-compromised too. Perturbing it could raise aneuploidy in healthy tissue — his predisposition mechanism.

ptc_triage.py — any nonsense variant, any recessive disease.

Next step is a dish, not a prescription.